

Name \_\_\_\_\_ Period \_\_\_\_\_

| <b>Skill 29.1 Exercise 1</b>                                                                                                                        |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|--|
| label each pair of events as <b>independent</b> or <b>dependent</b> and give a one-sentence reason                                                  |  |
| A coin is flipped twice. Event A: first flip is heads. Event B: second flip is heads.                                                               |  |
|                                                                                                                                                     |  |
| A card is drawn from a deck, <u>not</u> replaced, then a second card is drawn. Event A: first card is a heart. Event B: second card is a heart      |  |
|                                                                                                                                                     |  |
| A card is drawn, replaced, shuffled, then a second card is drawn. Event A: first card is a heart. Event B: second card is a heart.                  |  |
|                                                                                                                                                     |  |
| A spinner with equal sections is spun twice. Event A: first spin lands on blue. Event B: second spin lands on blue.                                 |  |
|                                                                                                                                                     |  |
| A bag has 4 green and 6 yellow marbles. Two marbles are drawn without replacement. Event A: first marble is green. Event B: second marble is green. |  |
|                                                                                                                                                     |  |

| <b>Skill 29.2 Exercise 1</b>                                                                                     |  |
|------------------------------------------------------------------------------------------------------------------|--|
| label each pair of events as <b>mutually exclusive</b> or <b>not mutually exclusive</b> and give a short reason. |  |
| Event A: roll an even number or<br>Event B: roll an odd number                                                   |  |
| Event A: roll a number greater than 4 or<br>Event B: roll an even number                                         |  |
| Event A: roll a 2 or<br>Event B: roll a number less than 4                                                       |  |
| Event A: roll a prime number or<br>Event B: roll an odd number                                                   |  |

Name \_\_\_\_\_ Period \_\_\_\_\_

|                                                                                                       |  |
|-------------------------------------------------------------------------------------------------------|--|
| <b>Skill 29.3 Exercise 1</b>                                                                          |  |
| Calculate the probability of rolling a prime number or rolling an odd number by completing the steps. |  |
| <b>Step 1:</b> Is this a mutually exclusive or not mutually exclusive?                                |  |
| <b>Step 2:</b> What is $P(A)$ ?                                                                       |  |
| <b>Step 3:</b> What is $P(B)$ ?                                                                       |  |
| <b>Step 4:</b> What is $P(A \text{ and } B)$ ?                                                        |  |
| <b>Step 5:</b> What is $P(A \text{ or } B)$ ?                                                         |  |

|                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------|
| <b>Skill 29.4 Exercise 1</b>                                                                                       |
| A bag contains <b>3 red marbles</b> and <b>2 blue marbles</b> . Two marbles are drawn <b>without replacement</b> . |
| Write a conditional probability expression (using notation) for each statement:                                    |
| a. The probability that the <b>second marble is red</b> , given that the <b>first marble is blue</b>               |
| b. The probability that the <b>second marble is blue</b> , given that the <b>first marble is red</b>               |
|                                                                                                                    |
| For each statement above, calculate the probability.                                                               |
|                                                                                                                    |

|                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------|
| <b>Skill 29.4 Exercise 2</b>                                                                                    |
| A bag contains <b>3 red marbles</b> and <b>2 blue marbles</b> . Two marbles are drawn <b>with replacement</b> . |
| Write a conditional probability expression (using notation) for each statement:                                 |
| a. The probability that the <b>second marble is red</b> , given that the <b>first marble is blue</b>            |
| b. The probability that the <b>second marble is blue</b> , given that the <b>first marble is red</b>            |
|                                                                                                                 |
| For each statement above, calculate the probability.                                                            |
|                                                                                                                 |

Name \_\_\_\_\_ Period \_\_\_\_\_

**Skill 29.5 Exercise 1**

A bag has 3 red and 2 blue marbles. Two marbles are drawn **without replacement**. Write an expression for: *The probability the first is red **and** the second is blue*

Compute marble probabilities (without replacement). Using the same bag (3 red, 2 blue), two draws without replacement.

Calculate:

- $P(\text{Blue 1st and Blue 2nd})$
- $P(\text{Blue 1st and Red 2nd})$

Compute the probabilities with replacement using the same bag.

Name \_\_\_\_\_ Period \_\_\_\_\_

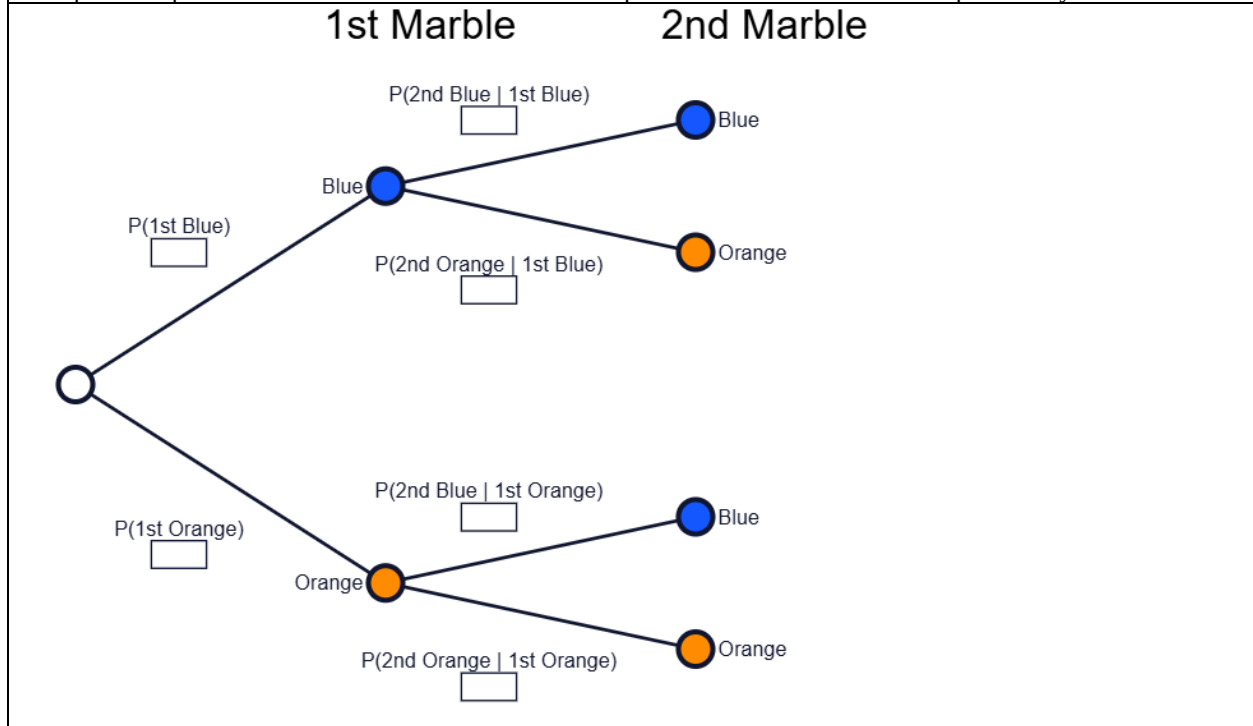
**Skill 29.6 Exercise 1**

Below is a two-draw probability tree for drawing marbles from a bag.

Given the distribution of blue and orange marbles below,



Complete the probabilities for each draw – assume no replacement. Then calculate the probability for ea



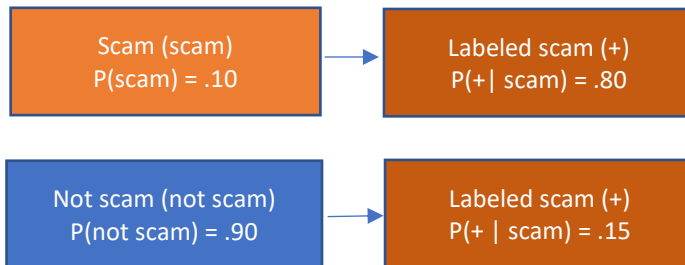
Name \_\_\_\_\_ Period \_\_\_\_\_

### Skill 29.7 Exercise 1

Your phone uses a filter to label text messages as “Scam” or “Not Scam.” The filter is helpful, but it is not perfect. Suppose the following are true,

- 10% of all messages are actually scams ( $P(\text{scam}) = .10$ )
- 90% of all messages are not scams ( $P(\text{not scam}) = .90$ )
- If a message is a scam, the filter labels it a scam 80% of the time ( $P(\text{label} | \text{scam}) = .80$ )
- If a message is not a scam, the filter still labels it as a scam 15% of the time ( $P(\text{label} | \text{not scam}) = .15$ )

These outcomes are shown visually below.



If your phone labels a message as “Scam,” what is the probability that the message is actually a scam? This is,  $P(\text{scam} | +)$

Consider 100 messages. Based on the probabilities above, how many are scams? How many are not scams?

Based on the probabilities above, how many of the *scams* are correctly labeled *scam*?

Based on the probabilities above, how many of the *not scams* are incorrectly labeled as *scam*?

What is the total messages that are labeled as *scam*?

What is  $P(\text{scam} | +)$ . That is, out of the messages labeled as scam (+), how many were actually scams?